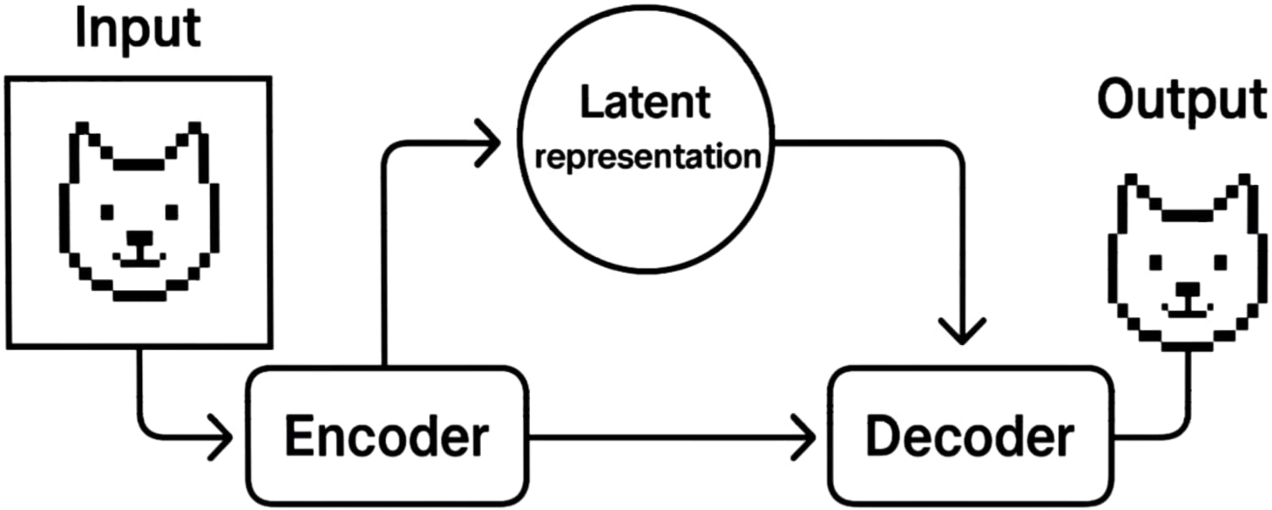
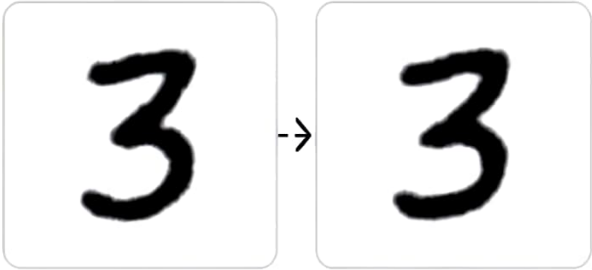
Basic Components of Variational Autoencoders (VAEs)



Understanding the VAE Loss Function

Reconstruction Loss



Reconstruction Loss = How different are these?

Trains the decoder to recreate inputs accurately.

KL Divergence

Learned Latent Distribution

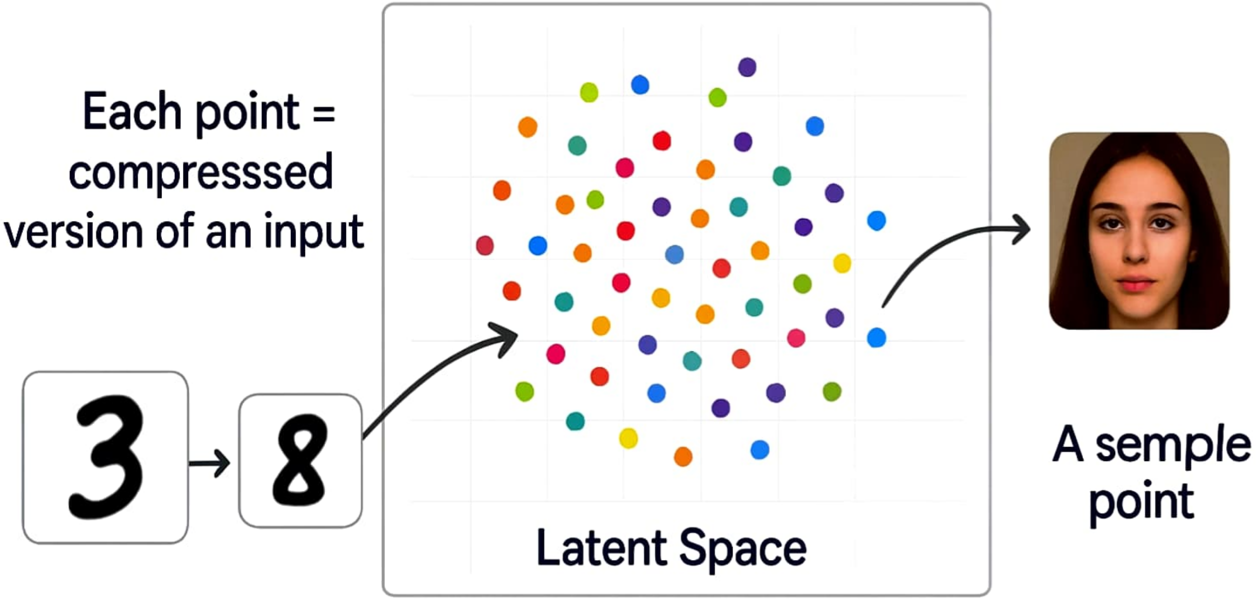
Standard Normal Distribution

KL Divergence = How far apart are these?

Encourages smooth, continuous latent space

Total loss = Reconstruction loss + KL divergence

Latent Space: The Heart of VAEs



Similar inputs cluster together. We can sample and interpolate!

Where Do We Use VAEs?

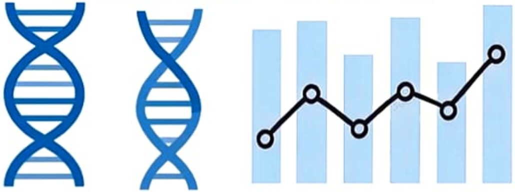
Image Generation

Creating new images that never existed.

Music/Audio

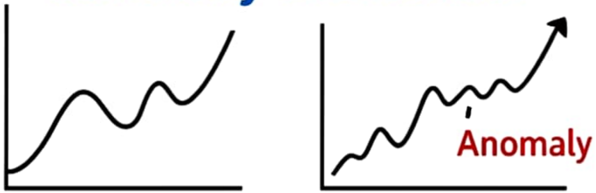
VAE-based sound synthesis.

Bioinformatics



Understanding gene patterns.

Anomaly Detection



Spotting unusual data behavior.